XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

EEE312 Wee6

Lab Practice

C Program to Compute the Eigenvalue and Eigenvector

Write a C program to solve any system of 4 linear equations using the *LU* decomposition based on the Cholesky's method. Display the matrix ***L*** and ***U*** and also the solutions to the linear equations at the end of the program.

.

```
// Beginning of the C Source code
```

```
1  // cholesky.c
2  // LU decomposition using the Cholesky's method.
3  #include <stdio.h>
4  #include <math.h>
5  #define N 4
6
7  void CholeskyLUdecomposition(double A[][N], double L[][N], double U[][N]);
8  void forwardSubstitution(double L[][N], double w[N], double b[N]);
9  void backSubstitution(double U[][N], double x[N], double w[N]);
0  void printMatrix(double [][N]); // print the NxN matrix
1  void printVector(double []); // print the Nx1 vector
```

Line 7: function Choelsky LU decomposition() to perform LU decomposition on matrix A to get matrix Land U.
Line 8: function forward Substitution() to perform forward substitution on matrix L with vector b to get vector w.
Line 9: function back Substitution() to perform back substitution on matrix U with vector w to get vector x.

Function printMatrix()

```
90  void printMatrix(double M[][N])
91  {
92      for(int i=0; i<N; i++)
93          for(int j=0; j<N; j++)
94          {
95              printf("%.4f\t\t", M[i][j]);
96              if(j==N-1) printf("\n");
97          }
98      printf("\n");
99  }
```

Line 95: Print the current element M[i][j] of matrix **M**.

Line 96: Print a newline character if the end of the row i is reached.

Function printVector ()

```
101 void printVector(double b[])
102 {
103     for(int i = 0; i<N; i++)
104     {
105         printf("%.4f\n", b[i]);
106     }
107     printf("\n");
108 }
```

Line 105: Print the current element b[i]of vector **b**.

Function CholeskyLUdecomposition ()

Line 44: Start from the first column on the left (j = 0), proceed to the next column on the right until the last column on the right (j=N-1).
Line 45: For every column j, start from the first element at the top (i= 0) and go to the next element below until the last element at the bottom (i= N-1).

```
41  void CholeskyLUdecomposition(double A[][N], double L[][N], double U[][N])
42  {
43      double sum = 0.0; // temporary variable to store the sum value
44      for(int j=0; j<N; j++) // column index j=0 to N-1
45          for(int i=0; i<N; i++) // row index i=0 to N-1
46      {
47          if(i==j) {// if diagonal element of L and U
48              sum=0.0;
49              for(int k=0; k<i; k++)
50                  sum += L[i][k]*L[i][k];
51              L[i][i] = sqrt(A[i][i]-sum);
52              U[i][i] = L[i][i];
53          }
```

Line 49-51: For the diagonal element *L*[*i*][*i*] implement the equation:

$$L[i][i] = \sqrt{A[i][i] - \sum_{k=0}^{i-1} (L[i][k])^2} \quad \text{for } i = 0, 1, 2, \ldots, N-1.$$

Line 52: Find the diagonal element *U*[*i*][*i*] = *L*[*i*][*i*].

```
54          else if(i>j){ // if BELOW the diagonal elements of L
55              sum = 0.0;
56              for(int k=0; k<j; k++)
57                  sum += L[j][k]*L[i][k];
58              L[i][j] = (A[i][j] - sum)/L[j][j];
59          }
60          else if(i<j) // if ABOVE the diagonal elements of U
61          {
62              U[i][j] = L[j][i];
63          }
64      }
65  }
```

Line 56-58: Find the element L[i][j] below the diagonal elements by using the equation

$$L[i][j] = \frac{A[i][j] - \sum_{k=0}^{j-1} (L[j][k] * L[i][k])}{L[j][j]} \quad \text{for } j = 0, 1, 2, \ldots, i-1.$$

Line 62: Find the element U[i][j] above the diagonal elements by using the equation U[i][j] = L[j][i].

Function backSubstidtution()

```
79  void backSubstitution(double U[][N], double x[N], double w[N])
80  {
81      double sum;
82      for(int i=N-1; i>=0; i--)
83      {
84          sum = 0.0;
85          for(int k=N-1; k>i; k--)
86              sum += U[i][k]*x[k];
87          x[i] = (w[i] - sum)/U[i][i];
88      }
89  }
```

Line 82: Start from the last row i= N□1, go backward until the first row i= 0.

Line 85 & 87: Implement the equation below to obtain x[i].

$$x[i] = \left[w[i] - \sum_{k=i+1}^{N-1} (U[i][k] \cdot x[k]) \right] \Big/ U[i][i]$$

Function backSubstidtution()

Line 70: Start from the first row i= 0, go forward until the last row i= N-1.

Line 73 & 75: Implement the equation below to obtain w[i].

$$w[i] = \left[b[i] - \sum_{k=0}^{i-1} (L[i][k] \cdot w[k]) \right] \Big/ L[i][i]$$

```
67  void forwardSubstitution(double L[][N], double w[], double b[])
68  {
69      double sum = 0.0;
70      for(int i=0; i<N; i++)
71      {
72          sum = 0.0;
73          for(int k=0; k<i; k++)
74              sum += L[i][k]*w[k];
75          w[i] = (b[i] - sum)/L[i][i];
76      }
77  }
```

Function main()

```
13  int main()
14  {
15      double A[N][N] = {{6,2,1,-1},{2,4,1,0},{1,1,4,-1},{-1,0,-1,3}}; //  coefficient matrix A
16      double b[N] = {2, -3, 1, -1}; //  matrix b
17      double x[N] = {0.0}; // unknown variables
18      double L[N][N] = {{0.0}}; // declare and initialize matrix L
19      double U[N][N] = {{0.0}}; // declare and initialize matrix U
20      double w[N] = {0.0}; // vector w
21
22      printf("The original matrix A is\n");
23      printMatrix(A); // print the original matrix A
24      printf("The original matrix b is\n");
25      printVector(b); // print the original matrix b
26      printf("\n");
27      CholeskyLUdecomposition(A, L, U);
28      forwardSubstitution(L, w, b); // forward substitution operation
29      backSubstitution(U, x, w); // back substitution operation
```

Line 15-20: Declare and initialize the matrices A, b, x, L, U and w.

Line 27: Perform LU decomposition on A to get Land U.

Line 28: Perform forwardsubstitution to get w.

Line 29: Perform backsubstitution to get x.

```
30      printf("The matrix L after the Cholesky's LU decomposition is\n");
31      printMatrix(L); // print the matrix L after LU decomposition operation
32      printf("The matrix U after the Cholesky's LU decomposition is\n");
33      printMatrix(U); // print the matrix U after LU decomposition operation
34      printf("The vector w after the Cholesky's LU decomposition is\n");
35      printVector(w); // print the matrix w after LU decomposition operation
36      printf("Solutions to linear equations are:\n");
37      printVector(x); // print the solved variables
38      return 0;
39  }
```

Line 30-37: Print out the results of the matrices L,U,w and x at the program output.

Output

```
The original matrix A is
6.0000          2.0000          1.0000          -1.0000
2.0000          4.0000          1.0000          0.0000
1.0000          1.0000          4.0000          -1.0000
-1.0000         0.0000          -1.0000         3.0000

The original matrix b is
2.0000
-3.0000
1.0000
-1.0000


The matrix L after the Cholesky's LU decomposition is
2.4495          0.0000          0.0000          0.0000
0.8165          1.8257          0.0000          0.0000
0.4082          0.3651          1.9235          0.0000
-0.4082         0.1826          -0.4679         1.6066

The matrix U after the Cholesky's LU decomposition is
2.4495          0.8165          0.4082          -0.4082
0.0000          1.8257          0.3651          0.1826
0.0000          0.0000          1.9235          -0.4679
0.0000          0.0000          0.0000          1.6066

The vector w after the Cholesky's LU decomposition is
0.8165
-2.0083
0.7278
0.0252

Solutions to linear equations are:
0.6649
-1.1780
0.3822
0.0157
```